

Vectorizing the Future: AstraDB's Role in Powering AI

Yasmeen Dao, Toto Kang, Emma Shek, Lilian Uong, Matthew Xue
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Abstract

DataStax Astra DB is, fittingly, a breakthrough in database technology designed for this emerging new field. This paper profiles the vector database capabilities of Astra DB, putting its pivot position as a key enabler for Generative AI applications to make their way from the concept phase into smooth production. Built on the battle-tested Apache Cassandra frame, Astra DB introduces a new way of dealing with mixed workloads, including vectors, non-vectors, and streaming data with top efficiency and precision. This allows concurrent updates and queries. That means developers are enabled to write Gen AI applications that are more responsive with less hallucination. With the real-time updating of vector data, comprehensive APIs, and data pipelines, Astra DB helps you grow AI-driven solutions in advanced domains like gaming, finance, and e-commerce.

The strengths in the Astra DB architecture, as surfaced by the paper, include the industry-leading vector database capabilities: improved relevance in vector search results, billions of vectors across high throughput, and minimal latency. The result is high scalability and flexibility of linear performance scaling with fluctuating workloads and rapid growth, both designed into the nature of Astra DB through Cassandra's distributed architecture. It also outlines how easy it is to integrate with modern applications and that Astra DB is a fully managed DBaaS, coming to the conclusion that there is no need for administrative overhead by developers. Although these are the many strengths it has, the paper also lists down its potential weaknesses: steep learning curves for the new users, a more expensive pricing model, limited options in terms of customization, and a relatively new community than others. With that, this paper seeks to highlight the potentiality of Astra DB as fuel for transformative Gen AI applications across industries, based on its comprehensive understanding of the features and what to consider before adoption.

Introduction

DataStax Astra DB emerges as a revolutionary vector database tailored specifically for developers venturing into the realm of Generative AI. With its robust capabilities, Astra DB facilitates a swift transition of Generative AI applications from the conceptual stage to full-fledged production. It offers a comprehensive suite of APIs, data pipelines, and an extensive range of ecosystem integrations, enabling the construction of Gen AI applications on a foundation of real-time data. This ensures the development of more precise AI models that can be seamlessly deployed in production environments. Built upon the resilient framework of Apache Cassandra, Astra DB distinguishes itself by accommodating vector data updates in real-time, thus catering to the most demanding data and streaming workloads with unparalleled security across any cloud platform. Vector search allows users to leverage the power of vector similarity calculations to search and retrieve for data. Vector search prioritizes on the similarity and meaning of semantics of data points rather than matching documents based on the occurrence of terms

DataStax designed Astra DB with a clear focus on handling mixed workloads with utmost efficiency. It adeptly manages a variety of data types, including vectors, non-vectors, and

streaming data, while ensuring concurrent updates and queries are executed with precision. The architecture of Astra DB, grounded in Apache Cassandra, enables it to perform simultaneous search and update operations on distributed data and streaming workloads with minimal latency. This capability is instrumental for developers aiming to create responsive and accurate Generative AI applications that minimize the occurrence of hallucinations.

Use-Cases

AstraDB can serve as a backend solution tailored for developers who are looking to implement Generative AI into their applications. Its infrastructure allows developers to swiftly construct apps leveraging real-time data to empower functionalities such as feed, chat, and vector AI search.

Gaming

When it comes to gaming platforms, much of their success relies on their data stack being able to adapt to massive scalability all while guaranteeing high responsiveness and availability across the globe. In this cut throat environment, AstraDB offers a suite of features tailored to meet the demanding requirements of modern gaming systems. Its unparalleled low read/write latencies at a massive scale ensure a real-time interactive gaming experience, crucial for keeping players engaged at all times.

One of AstraDB's key strengths lies in its support for Document API and multimodel flexibility, allowing it to seamlessly manage the wide variety of data generated and utilized within gaming ecosystems. From layering profile data and virtual goods catalogs to managing virtual currencies, player leaderboards, communities, and messaging tools, AstraDB provides a comprehensive solution for organizing and accessing diverse datasets with ease.

Moreover, AstraDB encourages gaming platforms to create real-time data pipelines for analytics, enabling developers to gain valuable insights into player behavior, trends, and performance metrics. These statistics can not only enhance the gaming experience but also enable data-driven decision-making for game optimization and content creation.

AstraDB is able to facilitate real-time updates to the game program itself, allowing for dynamic adjustments to in-game inventory, virtual goods, player achievements, leaderboards, and more. This keeps the gaming environment responsive and tailored to player interactions.

Its multimodal and globally distributed architecture further strengthens its appeal for gaming platforms, as it is able to seamlessly connect massive multiplayer online games across the world.

Finance

With the Financial Services sector trajecting towards a future of increased usage of AI technology and online services, AstraDB offers the leverage of servicing core transactions and analytics without compromising on response times or service availability.

AstraDB has the horsepower to handle core transactions at lightning-fast speed and reliability to guarantee that financial institutions can execute trades, process payments, and manage accounts without any disruptions or delays. Its high availability warrants that critical services remain accessible round-the-clock, providing customers with peace of mind and confidence in the stability of their financial transactions.

AstraDB has the functionality to derive valuable insights from vast amounts of data so that financial institutions can confidently make informed decisions, identify trends, and mitigate risks effectively. It can also provide assistance in detecting fraudulent activities, optimizing investment strategies, or personalizing customer experiences.

E-Commerce

In the dynamic landscape of e-commerce, customer expectations for seamless browsing experiences and efficient shopping transactions are the foundation of the industry. AstraDB's ability to effortlessly accommodate hundreds of thousands of queries with low read/write latencies ensures that product catalogs can be browsed and shopping carts managed with unparalleled speed and responsiveness. This scalability is essential for e-commerce platforms to cater to a large number of customers without compromising satisfaction or performance.

AstraDB's multi-model flexibility opens the opportunity for e-commerce platforms to manage a diverse array of data related to product catalogs, pricing, descriptions, and shopping carts efficiently. By supporting document APIs, AstraDB is able to simplify data management and access, enabling developers to build powerful and user-friendly browsing experiences.

E-commerce platforms can look forward to sending data instantly to distributed search engines, machine learning models, and data warehouses for near real time search and analytics, recommendations, and actionable insights with the use of AstraDB's seamless integration functionalities. This ensures that customers receive relevant and personalized recommendations based on their browsing history, preferences, and behaviors, enhancing their overall shopping experience.

With real-time updates to product availability, dynamic pricing, and shipping status, e-commerce platforms can have timely responses to changes in inventory, demand, and market conditions. This level of agility and responsiveness is crucial for maintaining customer trust and loyalty in a competitive e-commerce landscape.

Lastly, AstraDB's ability to span across multiple data centers or cloud regions enables e-commerce businesses to expand their operations internationally.

Tutorial

Purpose:

Vectors are properties of data as coordinates of dimensions with magnitude and direction. *Vector based searching* allows for the representation of data in multidimensional space and enables quick and effective assessment of approximating similarity in a generative AI approach. This can be helpful for companies in need of product recommendation systems in e-commerce. Similarity metrics can be used to compare product and user vectors to identify products that are similar to what users interact with.

Compatible Languages:

Python, Java, Node.js, C#, C++

Installation/Setup:

1. Go to AstraDB website and sign in to [Datastax](#)
2. Create Database cluster

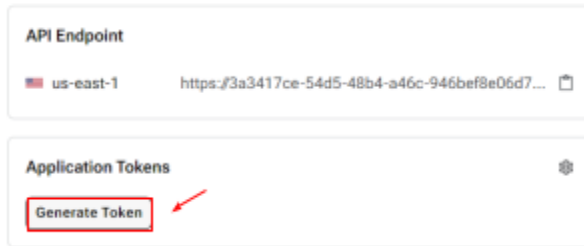
The image shows the AstraDB 'Quick Access' section with three buttons: 'Create a Database', 'Generate Token', and 'Invite your team'. The 'Create a Database' button is highlighted with a red box. Below this is the 'Create Database' modal window. It has a title bar with 'Create Database' and a close button. The modal is divided into two main sections: 'Select a deployment type' and 'Configuration'. Under 'Select a deployment type', there are two options: 'Serverless (Vector)' which is marked as 'Recommended' and described as 'An all-in-one database solution, optimized for Vector and Generative AI workloads', and 'Serverless (Non-Vector)' described as 'A more traditional database solution without any of our new vector capabilities'. The 'Serverless (Vector)' option is selected. The 'Configuration' section contains three fields: 'Database name*' with the value 'hw6_r6d', 'Provider*' with a dropdown menu showing 'Amazon Web Services', and 'Region*' with a dropdown menu showing 'us-east-1'. Below these fields is a grey bar with the text 'Upgrade to get more regions and premium features' and an 'Upgrade' button. At the bottom of the modal are two buttons: 'Cancel' and 'Create Database'.

Connection with Python Client:

1. `pip install --upgrade astrapy`

2. Generate application token in Astra Portal.

Database Details



3. Copy API Endpoint and auth token values from database details in Astra Portal and exported values to Pycharm or another application.

```
from astrapy.db import AstraDB

# initialize client
db =
AstraDB(token="AstraCS:heLjPGGrTBYDokfNjaGAqlygv:16bf8857f087526227c0f4cd112fb07a1e2d4b2a4
c1698ed721e29c77da465c6",
        api_endpoint="https://3a3417ce-54d5-48b4-a46c-946bef8e06d7-us-east-1.apps.astra.datastax.com",
        namespace=None
)

# test connection
print(f'Connected to Astra DB: {db.get_collections()}')
```

Performing Basic Operations (CRUD) and Queries:

Data API: Foundational vector API for Astra DB Serverless databases

1. *Create* collection.

```
# create collection
collection = db.create_collection("vec_test", dimension=4, metric="cosine")
print(collection)
```

Using `create_collection` specifies a new collection is created.

The default similarity metric is cosine, but other metrics include dot product and Euclidian distance. A dimension of four suggests that each document in the collection will contain vectors of length four. The metric and dimension are optional.

2. *Find* a collection.

```
collection = db.collection("vec_test")
```

Able to find one collection or iterate to find all the collections in the database.

3. *Insert* multiple documents into collection.

```
# insert documents in collection
documents = [
    {
        "_id": "1",
        "text": "Hello World",
```

```

    "$vector": [0.1, 0.2, 0.34, 0.45],
  },
  {
    "_id": "2",
    "text": "astra db",
    "$vector": [0.05, 0.08, 0.3, 0.6],
  },
]
res = collection.insert_many(documents)
print(res)

```

Can also insert one document at a time using `collection.insert_one` or even use chunk inserts to deal with large sets of data.

```

# assume documents represent a lot of data
result = collection.chunked_insert_many(
    documents=documents,
    chunk_size=20,
    concurrency=3, # insert 3 chunks at a time)

#Another example for text file and chunking into paragraphs
result = doc_chunker.chunk_file("filename")

```

4. Perform similarity search with `vector_find`.

```

# perform a similarity search
query = [0.15, 0.11, 0.12, 0.13]
results = collection.vector_find(query, limit=2, fields=["text", "$vector"])

for document in results:
    print(document)

```

Output:

```

{'status': {'insertedIds': ['2', '1']}}
{'_id': '1', 'text': 'Hello World', '$vector': [0.1, 0.2, 0.34, 0.45], '$similarity': 0.93756264}
{'_id': '2', 'text': 'astra db', '$vector': [0.05, 0.08, 0.3, 0.6], '$similarity': 0.87463874}
{'status': {'ok': 1}}

```

Can also find a single document using querying.

```

results = collection.find({
    "text": "astra db"})

```

Can also order results by similarity

```

results = collection.find(
    {"text": "astra db"},
    sort={"$vector": [0.05, 0.08, 0.3, 0.6]}
)

```

5. *Update* a document in a collection

```

update_result = collection.update_one({"_id": "1"}, {"$set": {"text": "New Text"}})

```

Output:

```

{'_id': '1', 'text': 'New Text', '$vector': [0.1, 0.2, 0.34, 0.45], '$similarity': 0.93756264}

```


Upsert to update a document if one exists, otherwise insert a new document

```
upsert= collection.upsert(  
document={"_id": "id1", "product_name": "old product"}  
)
```

6. *Delete* multiple documents in a collection.

```
res = collection.delete_collection(collection_name="test2")  
print(res)
```

Output:

Connected to Astra DB: {'status': {'collections': ['test', 'test1', 'test2', 'vec', 'vector_test']}}

Connected to Astra DB: {'status': {'collections': ['test', 'test1', 'vec', 'vector_test']}}

Or delete a single document based on a specific attribute in the collection.

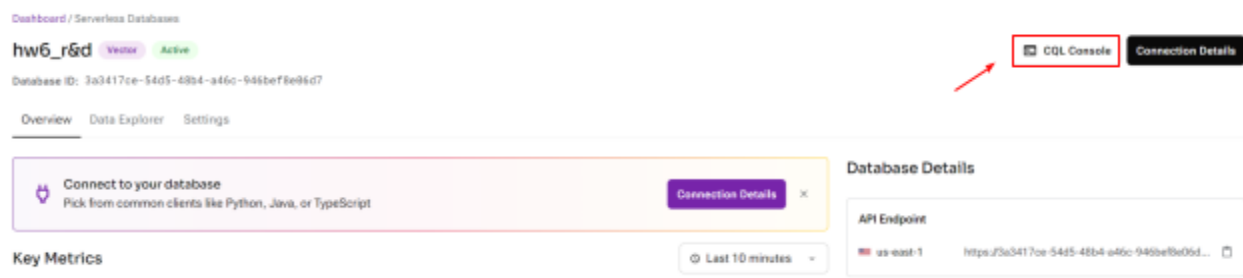
```
res = collection.delete_one(id="1")
```

Using CQL:

Astra DB Serverless leverages Apache Cassandra, therefore enabling the use of Cassandra Query Language (CQL) queries. You can use the standalone CQL shell or the embedded CQLshell in the Astra Portal that is used below.

Sample example: Recommendation engine in e-commerce using vector search. Helps suggest products based on browsing or purchasing history and utilizes vectors to approximate the nearest neighbor.

In the created database go to CQL Console in the top right corner.



Make sure to have created a keyspace in Data Explorer that can be used.

hw6_r&d

Vector

Active

Database ID: 3a3417ce-54d5-48b4-4

Overview

Data Explorer

Settings

Namespace ⓘ

What's this?

default_keyspace

default_keyspace

tutorial_keyspace

Choose the name of the keyspace you want to use.

Connected as dao.ya@northeastern.edu.

Connected to cndb at cassandra.ingress:9042.

[cqlsh 6.8.0 | Cassandra 4.0.0.6816 | CQL spec 3.4.5 | Native protocol v4 | TLS]

Use **HELP** for help.

token@cqlsh> use tutorial_keyspace;

With the key terms of the data, a matrix can represent whether each id contains one of the keywords or not.

	iphone	pro	max	charging	cable	Macbook	case	Windows	surface	Windex	cleaner	outdoor
pi185	1	1	1	1	0	0	0	0	0	0	0	0
pi293	1	1	1	1	0	1	0	0	0	0	0	0
pi012	0	1	0	0	0	1	0	0	0	0	1	0
pi011	0	1	0	0	0	0	0	1	1	0	0	0
pi583	0	0	0	0	0	0	0	1	0	1	1	1

Create a table that holds the attributes of the vectors.

Creating spatial indexes on vector data, AstraDb can retrieve spatial information based on relationships and proximity, enabling accurate spatial queries even with large datasets.

```
token@cqlsh:tutorial_keyspace> CREATE TABLE electronic_vectors (product_id TEXT PRIMARY KEY, product_name TEXT, product_vector vector<float, 12>);
token@cqlsh:tutorial_keyspace> CREATE CUSTOM INDEX ON electronic_vectors (product_vector) USING 'StorageAttachedIndex';
```

Insert rows into the table.

```
token@cqlsh:tutorial_keyspace> INSERT INTO electronic_vectors (product_id, product_name, product_vector)
... VALUES ('pi185','iPhone pro max charging cable',[1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0]);
token@cqlsh:tutorial_keyspace> INSERT INTO electronic_vectors (product_id, product_name,
```

```

product_vector)
... VALUES ('pi293','iPhone pro max charging case',[1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0]);
token@cqlsh:tutorial_keyspace> INSERT INTO electronic_vectors (product_id, product_name,
product_vector)
... VALUES ('pi012','Macbook pro case',[0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0]);
token@cqlsh:tutorial_keyspace> INSERT INTO electronic_vectors (product_id, product_name,
product_vector)
... VALUES ('pi011','Windows surface pro',[0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0]);
token@cqlsh:tutorial_keyspace> INSERT INTO electronic_vectors (product_id, product_name,
product_vector)
... VALUES ('pi583','Windex Outdoor Windows cleaner',[0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1]);
token@cqlsh:tutorial_keyspace> SELECT * FROM electronic_vectors;

```

Output:

product_id	product_name	product_vector
pi012	Macbook pro case	[0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0]
pi583	Windex Outdoor Windows cleaner	[0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1]
pi185	iPhone pro max charging cable	[1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0]
pi011	Windows surface pro	[0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0]
pi293	iPhone pro max charging case	[1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0]

(5 rows)

The query returns the similarity of products for the queried vector using the approximate nearest neighbor.

```

token@cqlsh:tutorial_keyspace> SELECT product_id, product_name,
similarity_cosine(product_vector, [1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0]) AS SIMILARITY FROM
electronic_vectors ORDER BY product_vector ANN OF [1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0] LIMIT 6;

```

Output:

product_id	product_name	similarity
pi185	iPhone pro max charging cable	1
pi293	iPhone pro max charging case	0.9
pi012	Macbook pro case	0.629099
pi011	Windows surface pro	0.629099
pi583	Windex Outdoor Windows cleaner	0.5

(5 rows)

Alternative method for loading data:

Use the built in AstraDB Data Loader

On the Astra DB dashboard navigate to the database you would like to load data. Select “Load Data.”

Dashboard / Serverless Databases

Nope Active

Overview Health Connect CQL Console CDC Settings

Load Data

Connect

Usage for Current Billing Period ⓘ

Read Requests

0

Write Requests

0

Storage Consumed ⓘ

423.67 KB

Data Transfer

86.02 MB

Once the data loader launches the following options appear:

1. Upload your own dataset.
2. Load an example dataset.
3. Load data from an Amazon S3 bucket that contains exported DynamoDB data.

Upload your own dataset:

Drag and drop personal .csv files into the Astra DB Data Loader. Columns must not contain spaces ex. Item_Cost. The status bar will indicate how much data has been uploaded.

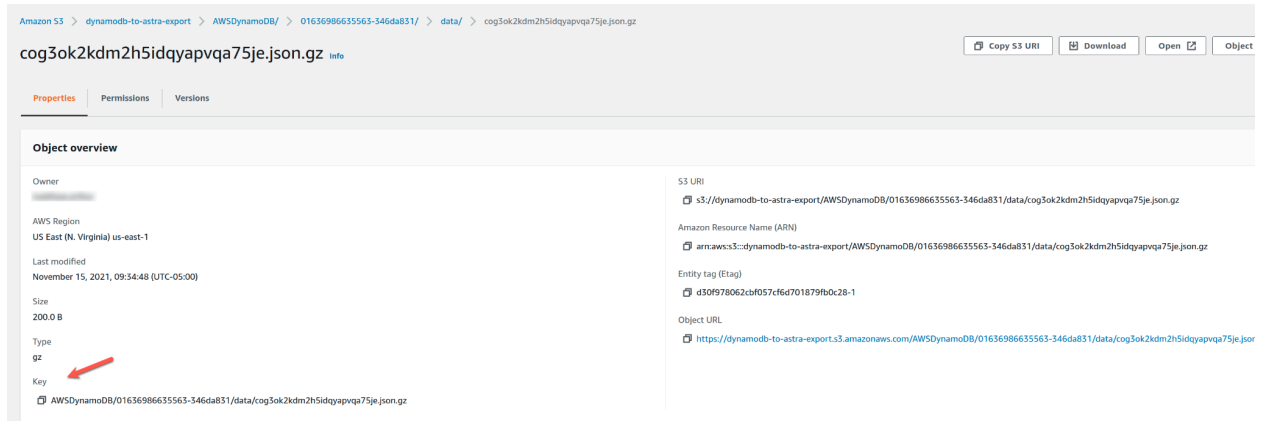
Load an example dataset:

There are two examples provided. Select one as the sample dataset.

Load data from an Amazon S3 bucket that contains exported DynamoDB data:

Export your DynamoDB data to S3. Grant read access to a specific role in the AWS console. Use the provided bucket policy to allow access to your S3 bucket. This policy lets Astra DB automation pull data from your shared S3 bucket and load it into Astra DB. You can remove this permission once the data load is complete.

Input your S3 Bucket name and the Key value. To find the Key, go to the S3 subdirectory in the AWS console where your DynamoDB data is exported. Look for the Key on the Properties tab.



Once the data loader option is specified, select Next.

Name the table for this dataset:

The data will appear in the **Data Previews & Types**

Astra DB Data Loader will automatically select the data type for each column. If needed, change this selection.

Select the partition key and clustering column for the data. Then select next.

Keys and Clustering

Partition Keys*

Search

▼

Clustering Columns

Search

▼

Back

Next

Select the database from the dropdown menu.

Select your keyspace from the available keyspace.

Select Next:

There will be confirmation that the data is being imported. The dataset will begin uploading into the database. There will be an email notifying the start and the completion of the data load. To interact with the data, **connect to your database**.

Terminology:

Approximate Nearest Neighbors(ANN) search - Quickly finds approximate nearest neighbors in large datasets

Chunking - Breaking text into subsets

Cluster - Two or more database instances that exchange messages

Collection - Datasets that contain forms of data such as vector embeddings

Vector - Ordered list of numbers

Vector database - Stores vectors

Vector index - Data structure used to store and query vectors for similarity or distance

Strengths and Weaknesses

Strengths

Industry-Leading Vector Database: Astra DB supports the handling of high-dimensional vector data. Unlike standalone vector databases that require stitching and moving data from system to system, Astra DB offers a natively unified database for the entire application. It also boasts excellent vector search capabilities, as it delivers up to 18% more relevant vector search results, up to 15x higher throughput, and ultra-low latency on billions of vectors. Furthermore, Astra DB is currently the only vector database that allows immediate, real-time vector updates to applications while also meeting enterprise-scale requirements. Overall, Astra DB is an industry-leading database solution and is largely recognized as such; it won the prestigious *TrustRadius*’ 2023 Top-Rated award in four different categories: Database-as-a-Service (DBaaS), Non-Relational Databases, NoSQL Databases, and Open-Source Databases.

Scalability: Astra DB has great horizontal scalability thanks to its inheritance of the distributed architecture of Apache Cassandra. This structure allows users to seamlessly meet increasing volumes of data and workloads by simply adding more nodes without large disruptions within the database. Astra DB’s horizontal scalability makes it a great fit for organizations that need a dynamic solution to fluctuating user demands or are facing rapid growth.

Furthermore, Astra DB’s performance scaling is largely linear, so as a user adds new nodes to the clusters in the database, a linear improvement in the overall system performance is exhibited. This empowers organizations to scale Astra DB clusters without worrying about spikes in traffic or sacrificing throughput and response times.

Astra DB is also scalable geographically, as it has a feature to deploy clusters in different regions or cloud data centers. Doing so allows Astra DB to localize data, reducing latency to a minimum while maximizing availability for users globally.

Flexibility: Astra DB does not tie developers down with strict, predefined schema definitions which allows for a large degree of flexibility and room for dynamic innovation. Developers can freely modify data types, add new fields, or define indexes without downtime or sacrifice in

ongoing database operations. Astra DB's flexibility also enables the handling of all levels of data structures, varying from vector, non-vector, unstructured, semi-structured, and fully structured data. Lastly, Astra DB excels in providing flexibility regarding resource allocation. It comes with features like auto-scaling which allows developers to set a resource utilization threshold in which the system would adjust resource usage depending on peak versus off-peak times.

Easy Integration: Astra DB is able to integrate well with many modern-day applications, tools, and frameworks. The driver and client supported by Astra DB enables developers to interact with Astra DB through many different programming languages. Moreover, Astra DB supports APIs such as CQL to Document (JSON), REST, GraphQL, and gRPC. Lastly, Astra DB is built on open-source technologies and can be used on any cloud provider or on-premises.

Fully-Managed Database Service: Astra DB is a database-as-a-service product that is fully managed by DataStax which minimizes the need for organizations to handle administrative chores such as maintenance, patches, backups, and upgrades. Users can focus their time on developing instead of managing.

Weaknesses

Steep Learning Curve: Astra DB is built on Apache Cassandra which has an industry perception of being complex. For those who are unfamiliar with Cassandra, Astra DB can be fairly difficult to use at first. An organization may have to go through a lot of additional training efforts to ensure users understand the architecture of a distributed data model. On the other hand, for an organization that is already using or has familiarity with Cassandra, the use of Astra DB will come very naturally. Overall, Astra DB could do a better job providing documentations, lessons, and onboarding material for new users.

Price: Astra DB is on the higher end for pricing compared to other similar solutions. The database's fully-managed service model is a likely explanation for the higher price compared to its self-managed competitors. The pricing structure of Astra DB is as follows: a free trial, a pay-as-you-go option, and an enterprise option. The pay-as-you-go option allows flexibility but makes it hard to predict and plan accordingly for long-term costs.

Insufficient Customization Availability: Due to Astra DB's fully-managed database service, there may be reduced flexibility and options compared to other self-managed competitors. So, an organization that requires specialized configurations and needs may find Astra DB to be limiting.

Limited Community: Astra DB is a fairly new database solution, having been released in 2020. Apache Cassandra itself has a robust community and ecosystem which is helpful to Astra DB users but the current community for the database itself is small. Expansion of community-contributed solutions would be beneficial to enhance the experience of Astra DB.

Conclusions and Summary

With the boom of AI, AstraDB has become very relevant and useful because of how effective and easy it is to implement Generative AI applications into production. AstraDB introduces a new way to manage mixed workloads like vectors, non-vectors, and streaming data with great accuracy and efficiency. AstraDB has a diverse array of use cases within the gaming, financial, and e-commerce industries because of its vector database, scalability, flexibility, and fully managed database service. AstraDB has many strengths, but there are also some challenges and weaknesses that this NoSQL database faces. These challenges include a steep learning curve for those who are unfamiliar with Apache Cassandra, high-end price compared to other similar solutions, a limited community of users, and there are limited specialized configurations that can be added. Despite some of these challenges, the benefits that AstraDB offers is an indispensable tool for developers who want to leverage the power of Generative AI.

As Generative AI continues to evolve and find new applications, the role of databases like Astra DB will become increasingly critical. By providing a reliable, scalable, and efficient database solution, Astra DB enables organizations to harness the full potential of Generative AI, driving innovation and transformation across industries. The future of database technologies, particularly in supporting the fast-growing field of Generative AI, looks promising, with Astra DB at the forefront of this revolution. It is anticipated that as Astra DB continues to mature and its community grows, it will overcome its current limitations and establish itself as the go-to database solution for Generative AI applications, furthering the frontier of what is possible in this exciting and rapidly evolving field.

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